THE RESPIRATORY BELLOWS

OPERATOR'S MANUAL: Ventilators, Chest Tubes, & Acid-Base Mechanics.

WARNING: PHOTOSENSITIVE ENVIRONMENT. ZERO WHITE LIGHT PERMITTED.

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CALIBRATION: THE RULE OF THE B'S

If the pH and the Bicarb are Both in the same direction, it is MetaBolic.

pH 7.30 (↓) & HCO3 20 (↓) = Metabolic Acidosis

pH 7.58 (↑) & HCO3 32 (↑) = Metabolic Alkalosis

⚠ pH 7.22 (↓) & HCO3 35 (↑) = Respiratory Acidosis ⚠

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SYSTEM OVERDRIVE: ALKALOSIS (pH > 7.45)

As the pH goes UP, so goes my patient.

- Irritability
- Hyper-reflexia (3+ & 4+)
- Tachypnea & Tachycardia
- Borborygmi (Increased bowel sounds)
- Seizure Risk (Action: Suction)

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SYSTEM SHUTDOWN: ACIDOSIS (pH < 7.35)

As the pH goes DOWN, so goes my patient.

- Hypo-reflexia (0 & 1+)
- Bradycardia & Bradypnea
- Lethargy / Obtunded / Coma
- Paralytic Ileus (Absent bowel sounds)
- Respiratory Arrest Risk (Action: Ambu-Bag)

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The Potassium Exception

Potassium moves OPPOSITE to pH.

ALKALOSIS (pH ↑) =
Potassium DOWN
(Hypokalemia)

ACIDOSIS (pH ↓) =
Potassium UP
(Hyperkalemia)

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EMERGENCY VENTING: MAC KUSSMAUL

The only acid-base imbalance to cause cause Kussmaul Respirations is Metabolic ACidosis.

MAC Kussmaul

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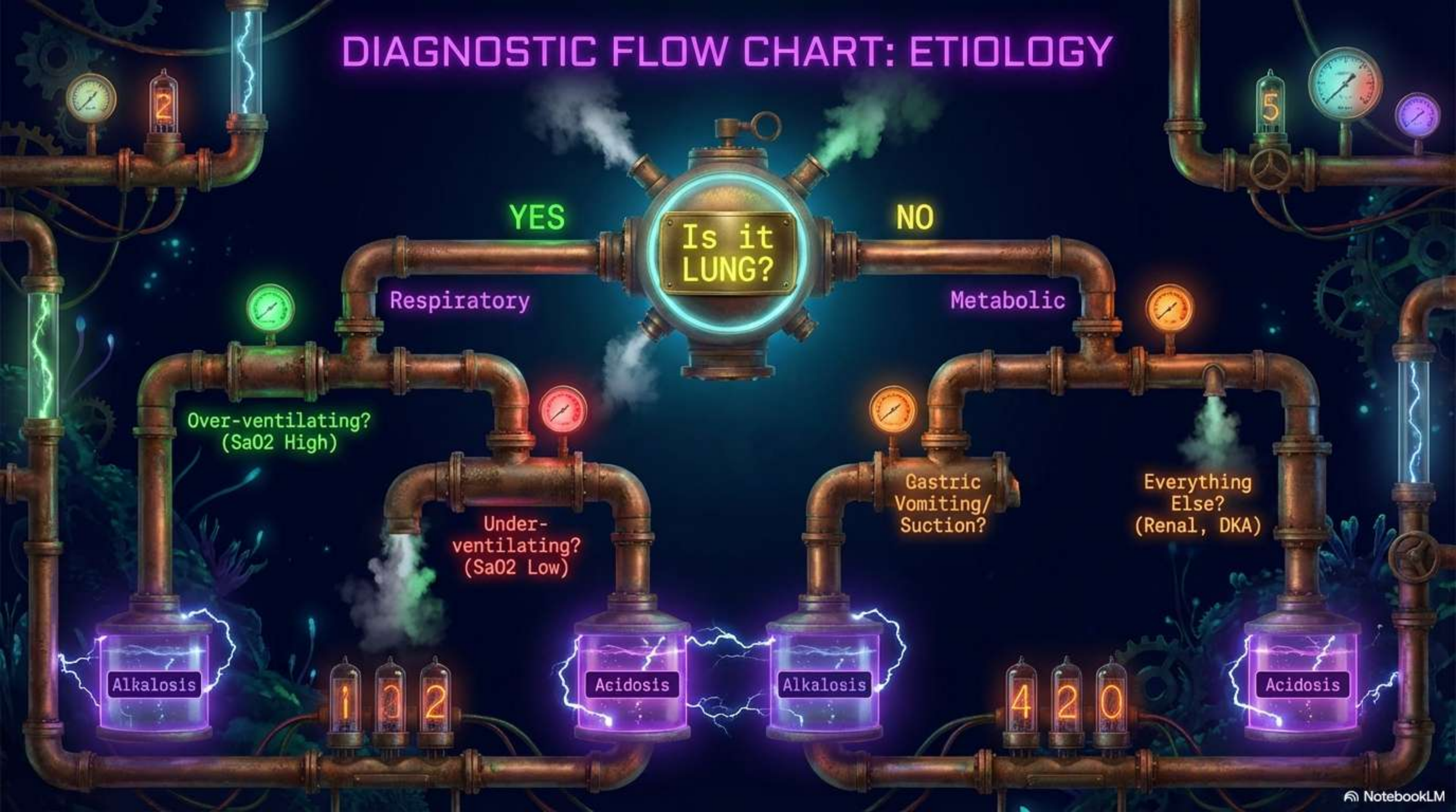
DIAGNOSTIC FLOW CHART: ETIOLOGY



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AUXILIARY PUMPS: VENTILATOR ALARMS

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HIGH PRESSURE ALARM: OBSTRUCTION

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Low Pressure Alarm: Disconnection

Causes

1. Main tubing disconnected
2. O2 sensor tubing disconnected

Interventions

- Reconnect the tubing.
- IF TUBE ON FLOOR: Bag the patient & Call Respiratory Therapy. (Never reconnect a dirty tube)

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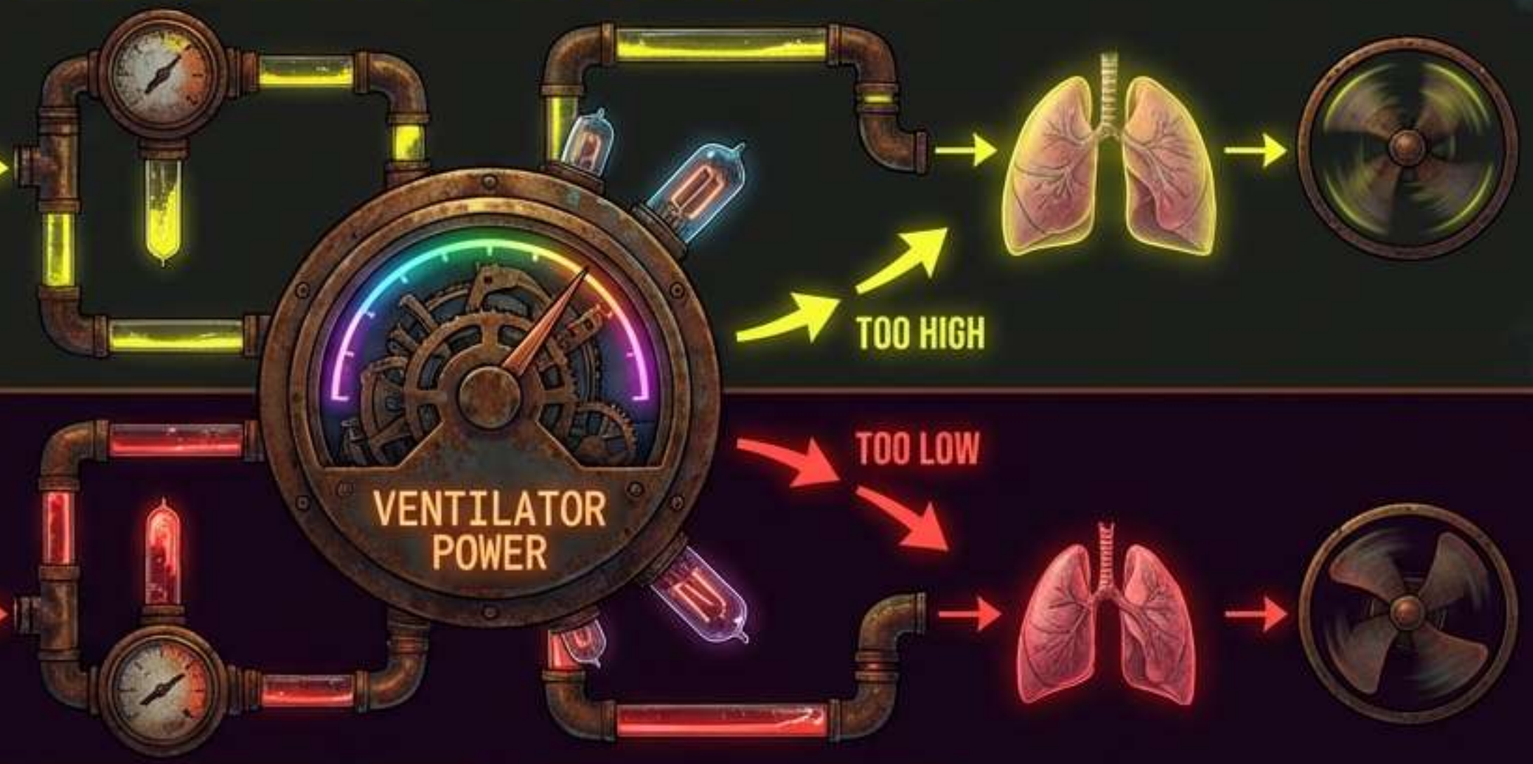
CALIBRATION: VENT SETTINGS VS. ABGs

RESPIRATORY ALKALOSIS (OVER-VENTILATING) -> SETTINGS ARE TOO HIGH.

pH > 7.45
PaCO2 < 35 mmHg
HCO3- NORMAL



pH < 7.35
PaCO2 > 45 mmHg
HCO3- NORMAL

ZONE B: RESPIRATORY ACIDOSIS (UNDER-VENTILATING) –> SETTINGS ARE TOO LOW.

WEANING PROTOCOL: DO NOT WEAN A PATIENT IN RESPIRATORY ACIDOSIS (UNDER-VENTILATED).

Action: Correct Acidosis First. Monitor ABGs Closely. Nursing Intervention Required.

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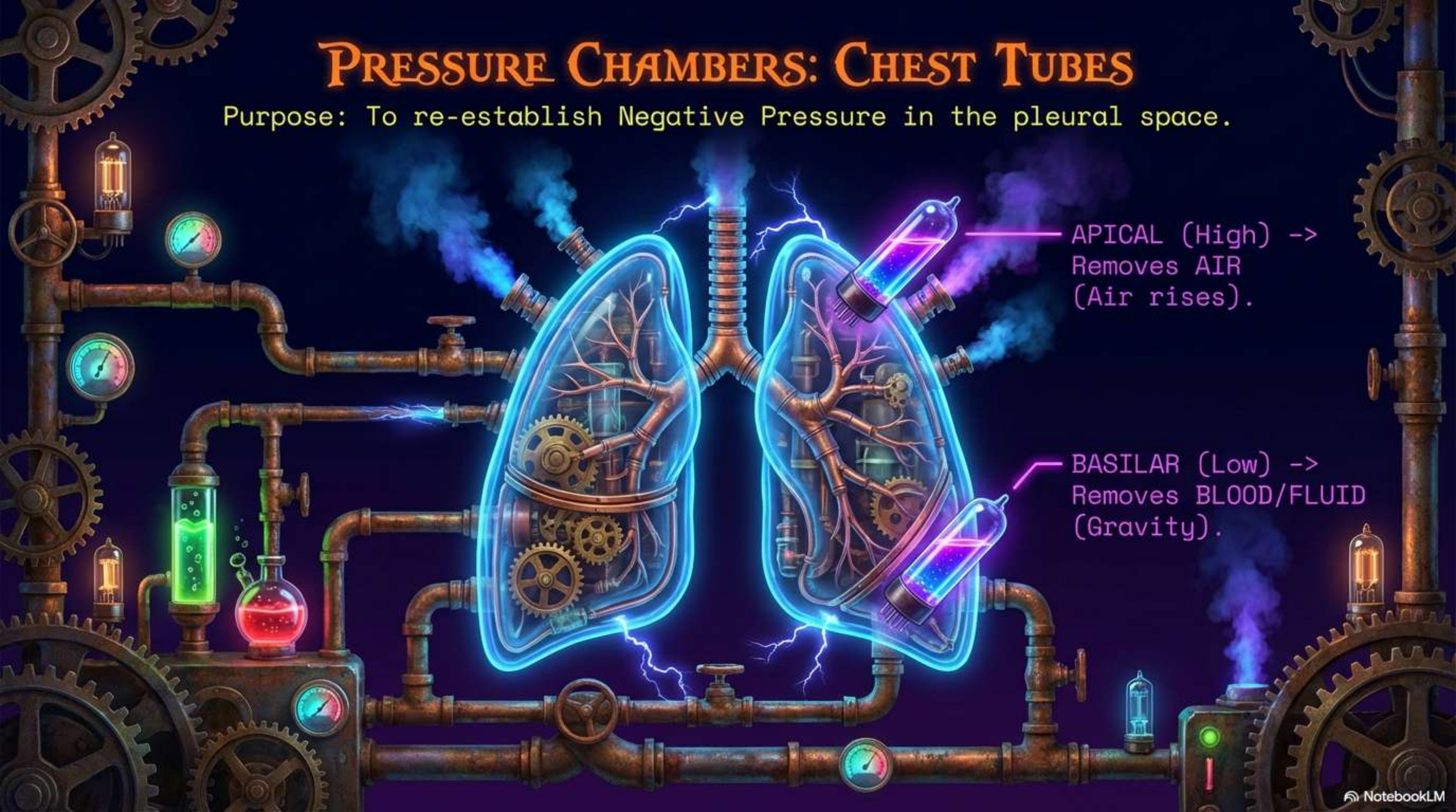
Pressure Chambers: Chest Tubes

Purpose: To re-establish Negative Pressure in the pleural space.



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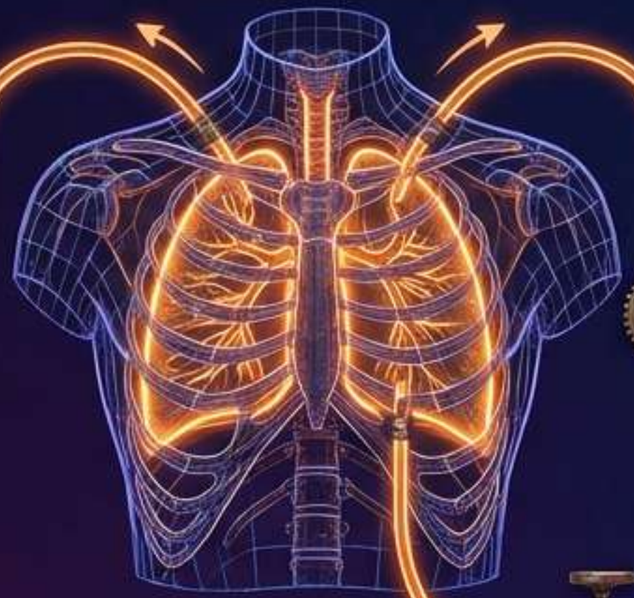
Configuration Protocols: Trauma

Unilateral Hemopneumothorax

Bilateral Pneumothorax



Post-Op Pneumonectomy

Reason: No lung = No pleural space

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EMERGENCY: WATER SEAL BROKEN

1. CLAMP (Max 15 seconds)
2. CUT (The tube away from broken device)
3. SUBMERGE (End of tube under sterile water)
4. UNCLAMP

PRIORITY: The BEST action is to SUBMERGE (Re-establishes the seal).

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EMERGENCY: TUBE PULLED OUT

FIRST ACTION:
Cover with Gloved Hand.

BEST ACTION:
Cover with Vaseline Gauze (Taped on 3 sides).

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THE BUBBLING MATRIX

WATER SEAL CHAMBER

Intermittent Bubbling = GOOD

Continuous Bubbling = BAD / LEAK

SUCTION CONTROL CHAMBER

Intermittent Bubbling = BAD / LOW SUCTION

Continuous Bubbling = GOOD

Rule: Sealed containers shouldn't bubble continuously.

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60
60
CRITICAL FAILURE:
THE "DEADLY Ds"
CO2 > 60
pO2 < 60
RESPIRATORY FAILURE

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DEADLY D PROTOCOL

Nursing Priorities

1. DO NOT LEAVE THE ROOM.
2. Prepare to Intubate & Ventilate.
3. Call Respiratory Therapy.
4. Call MD.

Contrast: If O_2 is 70-77 (Level C), just asseess & give O2.
If 60s (Level D), you must INTUBATE.

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PATHOLOGY: COPD / EMPHYSEMA

Air Trapping & Loss of Elasticity.

High CO_2 is expected/normal.

HYPOXIC DRIVE
(Patient relies on LOW O_2 to breathe).

Tx: Low Flow O_2 (Sat 88-92%). Pursed Lip Breathing.

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OBSTRUCTIVE PATHOLOGIES

PULMONARY EMBOLUS

V/Q Mismatch. Dead Space.

Sudden Hypoxia, Chest Pain.

PULMONARY EDEMA

Pink Frothy Sputum.

Diuretics (Furosemide).
Sit Upright (Legs Down).

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CHEMICAL ADDITIVE: AMINOPHYLLINE

Mechanism: Airway Anti-Spasmodic (Relaxes the muscle gears).

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